A Counterexample to the Group-Ring Determinant Conjecture

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Abstract

We disprove the unrestricted group-ring Determinant Conjecture. We construct a finitely generated group G and a square matrix over $\mathbb{Z}[G]$ that is invertible over $\mathbb{Q}[G]$ and has Fuglede–Kadison determinant strictly between zero and one. The logarithmic integral defining the determinant is finite. The construction rests on one cited input, Theorem 1.2.

1 Result, setting, and proof at a glance

A nonsingular integer matrix has determinant of absolute value at least one. The group-ring Determinant Conjecture asks for the analogue when the entries are finite integer combinations of elements of a discrete group and the determinant is that of Fuglede and Kadison [4].

Conventions. Let G be a discrete group and $\mathbb{Z}[G] \subseteq \mathbb{Q}[G] \subseteq \mathbb{C}[G]$ its group rings of finite sums. For $a = \sum_g a_g g$ write $[1]a = a_1$ for the identity coefficient and $a^* = \sum_g \overline{a_g}\, g^{-1}$; for matrices, $(A^*)_{ij} = (A_{ji})^*$. Let $(\delta_g)_{g\in G}$ be the standard orthonormal basis of $\ell^2(G)$ and $\lambda(g)\delta_h = \delta_{gh}$ the left regular representation, that is, $(\lambda(g)\xi)(h) = \xi(g^{-1}h)$. (The source writes L_g; we rename it because L_x later denotes a shear.) For $A \in \mathrm{M}_n(\mathbb{C}[G])$, the operator T_A acts on column vectors $\xi \in \ell^2(G)^n$ by

$$(T_A\xi)_i = \sum_{j=1}^{n} \sum_g (A_{ij})_g\, \lambda(g)\xi_j.$$

Since λ is a unitary representation, $A \mapsto T_A$ is a unital $*$-homomorphism: $T_{AB} = T_A T_B$ and $T_{A^*} = T_A^*$. Let $e_i \in \ell^2(G)^n$ have δ_1 in component i and 0 elsewhere. The *unnormalized* trace of a bounded operator T on $\ell^2(G)^n$ is

$$\tau_n(T) = \sum_{i=1}^{n} \langle Te_i, e_i\rangle, \qquad \text{so} \qquad \tau_n(T_A) = \sum_{i=1}^{n} [1]A_{ii}, \quad \tau_n(I_n) = n, \tag{1.1}$$

because $\langle \lambda(g)\delta_1, \delta_1\rangle$ is 1 for $g = 1$ and 0 otherwise. We abbreviate $\tau_n(A) = \tau_n(T_A)$.

The probability spectral measure of the positive square is

$$\mu_A(\Omega) = \frac{1}{n} \sum_{i=1}^{n} \langle \mathbf{1}_\Omega(T_A^*T_A)e_i, e_i\rangle \quad (\Omega \subseteq [0,\infty) \text{ Borel}),$$

and the spectral theorem gives $\tau_n(f(T_A^*T_A)) = n \int f\, d\mu_A$ for bounded Borel f. The square-matrix instance of the conjecture asserts that for every G, n and $A \in \mathrm{M}_n(\mathbb{Z}[G])$,

$$\int_{(0,\infty)} \log t\, d\mu_A(t) \geq 0. \tag{1.2}$$

The zero atom is omitted and the integral may be $-\infty$. The usual conjecture also permits rectangular matrices [7, Conjecture 13.2]; a square counterexample already contradicts it.

If T_A is invertible, then $\|T_A^{-1}\|^{-2} I \le T_A^* T_A \le \|T_A\|^2 I$, so μ_A is carried by a compact subinterval of $(0,\infty)$, on which log is bounded. With $\mathcal{N}(G)$ the group von Neumann algebra, the determinant convention is

$$\log \det_{\mathcal{N}(G)}(T_A) = \frac{1}{2}\tau_n\big(\log(T_A^* T_A)\big) = \frac{n}{2}\int_{(0,\infty)} \log t \, d\mu_A(t). \tag{1.3}$$

For trivial G this is $\log|\det A|$.

Relation to right multiplication (not used below). The literature lets matrices act on row vectors from the right. Let ι be the unitary $\delta_g \mapsto \delta_{g^{-1}}$ in each component. Since $\iota\lambda(g)\iota\,\delta_h = \delta_{hg^{-1}}$, for integral A the operator $\iota T_A \iota$ is $\xi \mapsto \xi A^*$ (row vector times matrix). As $\iota e_i = e_i$, both operators have the same measure μ_A; so (1.2) for A is the right-multiplication statement for A^*.

Theorem 1.1. *There are a finitely generated discrete group G_D, an integer $n \ge 1$, and $A \in \mathrm{M}_n(\mathbb{Z}[G_D])$ invertible over $\mathbb{Q}[G_D]$ such that*

$$0 < \det_{\mathcal{N}(G_D)}(T_A) < 1.$$

The integral in Equation (1.2) is finite and strictly negative. Thus the unrestricted group-ring Determinant Conjecture has a negative resolution.

A unital ring is *directly finite* if $ab = 1$ implies $ba = 1$. The proof uses one external result.

Theorem 1.2 ([8, Theorem 1.1])**.** *There exist a finite field K_2 of characteristic two, a finitely generated group H_2, and finite sums $a_2, b_2 \in K_2[H_2]$ with $a_2 b_2 = 1$ and $b_2 a_2 \ne 1$.*

Unverified step. Theorem 1.2 is quoted from the companion paper and is neither proved nor checked here; the source gives no explicit K_2, H_2, a_2, b_2 to test. It says the group algebra $K_2[H_2]$ is not directly finite, contrary to Kaplansky's direct-finiteness conjecture, so what the present text establishes is the implication "Theorem 1.2 $\Rightarrow$ Theorem 1.1".

According to the source, the companion (its Section 6) specifies the field, group presentation and sums by a terminating finite prescription and uses torsion; only the existence statement is used here. The field K_2 is not asserted to be $\mathbb{F}_2$ (hence Lemma 2.1), and a result only in odd characteristic would not supply the binary premise. Everything after Theorem 1.2 is proved below.

Context. Determinant bounds belong to the theory of L^2-invariants [7]. The bound (1.2) is known for large classes of groups, including amenable groups [9, Theorem 1.21, with its erratum] and sofic groups [3, Theorem 5]. It is distinct from determinant *approximation* by finite quotients: Kammeyer's counterexample [5, Theorem 2 and Section 2] has determinant 2 over the discrete Heisenberg group against at most $5^{1/3}$ on its congruence quotients of levels 3^i, which does not violate the lower bound one. The commuting projections of Section 2 are related to those of [2, Section 3] and [6, Section 2]; the analytic step is the tracial logarithmic derivative [4, Lemma 2], also in [1, Lemma 1 and Proposition 2(d)]. Both are proved here; nothing is imported from these works.

Proof at a glance. *Key idea.* Suppose a rational orthogonal projection P of size m has odd denominators and its reduction modulo 2 factors through r columns. Then for suitable $q = 2^k$ there is an integer matrix A that differs only by two block shears from an operator multiplying r coordinates by q and the range of P by $-q^{-1}$. Shears do not change the determinant, so $\log\det T_A = (r - \tau_m(P))\log q$, which is negative once $\tau_m(P) > r$. (For trivial G, $\tau_m(P) = \operatorname{rank} P = \operatorname{rank} \bar{P} \le r$, so nothing goes wrong, as expected.) A failure of direct finiteness in characteristic two yields such a P with $r = d$ and $\tau_m(P) = d + \theta$, $\theta > 0$.

1. Lemma 2.1: Theorem 1.2 gives X, Y over $\mathbb{F}_2[H_2]$ with $XY = I_d \neq YX$.
2. Proposition 2.2 (uses 1): over $G_D = (\bigoplus_{h\in H_2} C_3) \rtimes H_2$ there is a projection P of size $d+1$ with trace $d+\theta$ whose reduction factors through d columns.
3. Lemma 3.1: a factorization modulo 2 lifts to one modulo $q = 2^k$.
4. Lemma 3.2 (uses 3): the integral matrix A and its rational inverse.
5. Lemmas 4.1 and 4.2: shears preserve $\tau(\log(M^*M))$.
6. Proposition 4.3 (uses 4, 5): $\log\det T_A = (r - \tau_m(P))\log q$.
7. Section 5: items 2 and 6 give $\det T_A = q^{-\theta}$.

Items 3–6 hold for every group and do not depend on items 1–2.

2 From the scalar defect to a projection

All group-ring sums are finite and matrix products keep the indicated order.

Lemma 2.1. *Let K_2 be a finite field of characteristic two and H_2 a group. For $d = [K_2 : \mathbb{F}_2]$, there is an injective unital ring homomorphism*

$$\rho : K_2[H_2] \longrightarrow \mathrm{M}_d(\mathbb{F}_2[H_2]).$$

For $a_2, b_2 \in K_2[H_2]$ with $a_2b_2 = 1 \neq b_2a_2$, their images X, Y satisfy $XY = I_d$ and $C := I_d - YX \neq 0$, with

$$C^2 = C, \qquad XC = 0, \qquad CY = 0. \tag{2.1}$$

This turns the input into the hypothesis of Proposition 2.2.

Proof. Step 1. Choose an $\mathbb{F}_2$-basis of K_2 and let $m_\kappa \in \mathrm{M}_d(\mathbb{F}_2)$ be the matrix of $x \mapsto \kappa x$. Then $\kappa \mapsto m_\kappa$ is a unital ring homomorphism, injective because m_κ applied to $1 \in K_2$ returns κ.

Step 2. Let $\rho(\sum_h \kappa_h h)$ be the matrix with (i,j) entry $\sum_h (m_{\kappa_h})_{ij}\, h$. It is additive and unital. For monomials, the (i,j) entry of $\rho(\kappa g)\rho(\kappa' h)$ is $\sum_l (m_\kappa)_{il}(m_{\kappa'})_{lj}\, gh = (m_{\kappa\kappa'})_{ij}\, gh$, since elements of $\mathbb{F}_2$ commute with group elements; this is the entry of $\rho(\kappa\kappa' gh)$. So ρ is multiplicative. If $\rho(a) = 0$, every m_{κ_h} vanishes, hence every $\kappa_h = 0$ by Step 1.

Step 3. Put $X = \rho(a_2)$, $Y = \rho(b_2)$. Then $XY = \rho(1) = I_d$, and $YX = \rho(b_2a_2) \neq I_d$ by injectivity. From $(YX)^2 = Y(XY)X = YX$,

$$C^2 = I_d - 2YX + (YX)^2 = I_d - YX = C, \quad XC = X - (XY)X = 0, \quad CY = Y - Y(XY) = 0. \ \square$$

Proposition 2.2. *Let $X, Y \in \mathrm{M}_d(\mathbb{F}_2[H_2])$, $d \ge 1$, satisfy $XY = I_d \neq YX$. In the group $G_D = (\bigoplus_{h\in H_2} C_3) \rtimes H_2$, where H_2 acts by left translation on coordinate labels, there is an orthogonal projection $P \in \mathrm{M}_{d+1}(\mathbb{Z}[1/3][G_D])$ and a finite set $S \subseteq H_2 \setminus \{1\}$ such that*

$$\tau_{d+1}(P) = d + \theta, \qquad \theta = \frac{1}{3}\left(\frac{2}{3}\right)^{|S|} > 0, \qquad \bar{P} = uv,$$

where u has size $(d+1)\times d$ and v has size $d\times(d+1)$ over $\mathbb{F}_2[G_D]$. The integer $D=3^{1+|S|}$ clears all coefficients of P.

Here "orthogonal projection" means $P=P^*=P^2$, and $\bar{P}$ is the reduction modulo 2 (Step 5). The proposition is combined with Proposition 4.3 in Section 5.

Proof. Plan. A rational idempotent p built from the C_3 coordinates isolates one coefficient of $C=I_d-YX$. It carries the extra trace θ, while modulo 2 the defect C absorbs it into a factorization through d columns.

Notation. Let z_h generate the h-th copy of C_3 and $\Gamma=\bigoplus_h C_3$, an abelian normal subgroup of G_D. In G_D we have $hz_kh^{-1}=z_{hk}$, and $H_2\le G_D$, so $\mathbb{F}_2[H_2]\subseteq\mathbb{F}_2[G_D]$. The matrix C is nonzero and satisfies (2.1): Step 3 of Lemma 2.1 used only $XY=I_d$.

Step 1 (normalizing one entry). Choose i,j and $h_0\in H_2$ such that h_0 has nonzero coefficient in C_{ij}; that coefficient is 1, the field being $\mathbb{F}_2$. Hence $h_0^{-1}C_{ij}$ has identity coefficient 1, and

$$h_0^{-1}C_{ij}=1+\sum_{s\in S}s \tag{2.2}$$

for a finite $S\subseteq H_2\setminus\{1\}$, possibly empty.

Step 2 (the idempotent p). Put

$$p_h=\frac{1+z_h+z_h^2}{3},\qquad p=p_1\prod_{s\in S}(1-p_s).$$

Since $z_h^3=1$, $(1+z_h+z_h^2)^2=3(1+z_h+z_h^2)$, and $z_h^*=z_h^2$; so $p_h=p_h^*=p_h^2$. All p_h lie in the commutative ring $\mathbb{Q}[\Gamma]$. Hence each $1-p_s$ is a self-adjoint idempotent, and so is the commuting product p. As $3p_1$ and $3(1-p_s)$ are integral, $3^{1+|S|}p$ is integral. (Compare [2, Lemmas 3.1 and 3.4].)

Step 3 ($psp=0$ for $s\in S$). From $sz_hs^{-1}=z_{sh}$ we get $sp_hs^{-1}=p_{sh}$, so

$$sps^{-1}=p_s\prod_{t\in S}(1-p_{st}).$$

This contains the factor p_s, while p contains $1-p_s$. All factors commute and $(1-p_s)p_s=0$, so $p\,(sps^{-1})=0$ and

$$psp=p(sps^{-1})s=0. \tag{2.3}$$

Only these two factors are used; the label sets $\{1\}\cup S$ and $s(\{1\}\cup S)$ may otherwise overlap, and inverse pairs or finite-order elements in S change nothing.

Step 4 ($[1]p=\theta$). The factors p_1 and $1-p_s$ ($s\in S$) are supported in the coordinates 1 and s, which are pairwise distinct because $1\notin S$. A product of elements of distinct coordinates of Γ is the identity only if every factor is. So the identity coefficient of p is the product of those of its factors, namely $[1]p_1=\frac{1}{3}$ and $[1](1-p_s)=\frac{2}{3}$. By (1.1),

$$\tau_1(p)=[1]p=\frac{1}{3}\prod_{s\in S}\frac{2}{3}=\theta>0. \tag{2.4}$$

Step 5 (reduction modulo two). Since 3 is invertible in $\mathbb{F}_2$, there is a ring homomorphism $\mathbb{Z}[1/3]\to\mathbb{F}_2$; applied to coefficients it gives a ring homomorphism $a\mapsto\bar{a}$ into $\mathbb{F}_2[G_D]$, and likewise on matrices. Thus $\bar{p}^2=\bar{p}$ and $\bar{p}s\bar{p}=0$ by (2.3). Multiplying (2.2) by $\bar{p}$ on both sides,

$$\bar{p}\,h_0^{-1}C_{ij}\,\bar{p}=\bar{p}^2+\sum_{s\in S}\bar{p}s\bar{p}=\bar{p}.$$

Step 6 (factorization). Let f be the row of length d with entry $\bar{p}h_0^{-1}$ at position i and zeros elsewhere, and w the column with entry $\bar{p}$ at position j and zeros elsewhere. Then $fCw = \bar{p}h_0^{-1}C_{ij}\bar{p} = \bar{p}$. Define

$$P = \mathrm{diag}(I_d, p), \qquad u = \begin{pmatrix} X \\ fC \end{pmatrix}, \qquad v = \begin{pmatrix} Y & Cw \end{pmatrix}.$$

Using $XY = I_d$, $XC = 0$, $CY = 0$, $C^2 = C$ in turn,

$$uv = \begin{pmatrix} XY & XCw \\ fCY & fC^2w \end{pmatrix} = \begin{pmatrix} I_d & 0 \\ 0 & fCw \end{pmatrix} = \mathrm{diag}(I_d, \bar{p}) = \bar{P}.$$

By Step 2, $P = P^* = P^2$ and $D = 3^{1+|S|}$ clears its coefficients; by (1.1) and (2.4), $\tau_{d+1}(P) = d+\theta$. □

3 Lifting the factorization and preserving integrality

From here to the end of Section 4, G is any group. If $D > 0$ is odd and $k \geq 1$, then D is a unit in $\mathbb{Z}/2^k\mathbb{Z}$, so reduction of integers extends to a ring homomorphism $\mathbb{Z}[1/D] \to \mathbb{Z}/2^k\mathbb{Z}$. Applied to coefficients it is a ring homomorphism on matrices over $\mathbb{Z}[1/D][G]$; for $k = 1$ it is written $^-$.

Why the matrix below. For $q = 2^k > 1$ and a projection P of size m put

$$J = -q^{-1}P - (I_m - P), \qquad B = \mathrm{diag}(qI_r, J).$$

The absolute value of B expands r coordinates by q and contracts the range of P by q^{-1}, which suggests $\log\det = (r - \tau_m(P))\log q$; but B is not integral. A block matrix with upper-left block qI_r and off-diagonal blocks V (upper), U (lower) has Schur complement equal to its lower-right block minus UV/q. For this to be J, the lower-right block must be $UV/q + J = q^{-1}(UV - P - q(I_m - P))$, which is integral exactly when $UV \equiv P + q(I_m - P)$ modulo q. The minus signs in J produce the difference $UV - P$; Lemma 3.1 supplies U, V.

Lemma 3.1. *Let G be a group, $m, r \geq 1$, D an odd positive integer, and $P \in \mathrm{M}_m(\mathbb{Q}[G])$ satisfy $P^2 = P$ and $DP \in \mathrm{M}_m(\mathbb{Z}[G])$. Suppose $\bar{P} = uv$ over $\mathbb{F}_2[G]$, where u and v have respective sizes $m \times r$ and $r \times m$. For $q = 2^k$, $k \geq 1$, the reduction E of P modulo q has a factorization $E = u_q v_q$ of the same sizes over $(\mathbb{Z}/q\mathbb{Z})[G]$.*

(The source calls this reduction $e = P_q$; we write E because e_i are basis vectors.) The lemma feeds Lemma 3.2.

Proof. Plan. Lift u, v arbitrarily; the lifted product is correct modulo 2 inside the corner cut out by E, and the error is removed by a finite geometric series.

Step 1. Put $\mathcal{R} = (\mathbb{Z}/q\mathbb{Z})[G]$. Reduction is a ring homomorphism, so $E^2 = E$; and the image of E under $\mathcal{R} \to \mathbb{F}_2[G]$ is $\bar{P}$.

Step 2. Choose lifts u_0, v_0 of u, v over $\mathcal{R}$, and set

$$Z = Eu_0v_0E, \qquad W = E - Z.$$

Both lie in the corner ring $E\,\mathrm{M}_m(\mathcal{R})E$, whose identity is E: $EW = W = WE$.

Step 3 ($W^k = 0$). Modulo 2, Z becomes $\bar{P}u\bar{v}\bar{P} = \bar{P}^3 = \bar{P}$, the image of E. So every coefficient of every entry of W is even. An entry of W^k is a sum of products of k entries of W, so its coefficients are divisible by $2^k = q$ and vanish in $\mathbb{Z}/q\mathbb{Z}$.

Step 4 (corner inverse). Let $Q = E + W + \cdots + W^{k-1}$ (so $Q = E$ if $k = 1$). Since $QE = Q$ and $Z = E - W$, telescoping gives $QZ = Q - QW = E - W^k = E$.

Step 5. Set $u_q = QEu_0$ and $v_q = v_0E$. Then $u_qv_q = Q(Eu_0v_0E) = QZ = E$. Only finite sums and products occurred, so supports stay finite. □

Choice of q. Since 2 is a unit modulo the odd integer D, some $k \geq 1$ has $2^k \equiv 1 \pmod{D}$ (any $k \geq 1$ if $D = 1$). Fix $q = 2^k > 1$ with $q \equiv 1 \pmod{D}$.

Lemma 3.2. *Suppose $P = P^* = P^2 \in \mathrm{M}_m(\mathbb{Q}[G])$, an odd $D > 0$ clears its denominators, and $\bar{P} = uv$ factors through $r \geq 1$ columns. Let $q = 2^k > 1$ satisfy $q \equiv 1 \pmod{D}$. Choose arbitrary lifts U, V over $\mathbb{Z}[G]$ of the factors u_q, v_q supplied by Lemma 3.1, so that UV reduces to E modulo q. Then*

$$A = \begin{pmatrix} qI_r & V \\ U & q^{-1}(UV - P - q(I_m - P)) \end{pmatrix} \in \mathrm{M}_{r+m}(\mathbb{Z}[G]) \tag{3.1}$$

has an inverse in $\mathrm{M}_{r+m}(\mathbb{Q}[G])$.

This is the matrix of Theorem 1.1; Equation (3.2) below is reused in Proposition 4.3.

Proof. Step 1 (integrality). Write $P + q(I_m - P) = qI_m + \frac{1-q}{D}DP$; it is integral because $D \mid q - 1$ and DP is integral. Its reduction modulo q is $0 + (1 - 0)E = E$, which is also the reduction of UV. So every coefficient of $UV - P - q(I_m - P)$ is an integer divisible by q, and A is integral.

Step 2 (shear factorization). With J, B as above, $P(I_m - P) = 0$ gives $J(-qP - (I_m - P)) = P + (I_m - P) = I_m$, so $J^{-1} = -qP - (I_m - P)$ and $B^{-1} = \mathrm{diag}(q^{-1}I_r, J^{-1})$. For $x \in [0, 1]$ put

$$L_x = \begin{pmatrix} I_r & 0 \\ xU/q & I_m \end{pmatrix}, \qquad R_x = \begin{pmatrix} I_r & xV/q \\ 0 & I_m \end{pmatrix}.$$

Then $L_xB = \begin{pmatrix} qI_r & 0 \\ xU & J \end{pmatrix}$, and multiplying by R_x on the right,

$$M(x) := L_xBR_x = \begin{pmatrix} qI_r & xV \\ xU & x^2UV/q + J \end{pmatrix}. \tag{3.2}$$

Since $UV/q + J = q^{-1}(UV - P - q(I_m - P))$, we get $A = M(1)$.

Step 3 (inverse). The inverses of L_x, R_x are obtained by negating the off-diagonal block. Hence $M(x)^{-1} = R_x^{-1}B^{-1}L_x^{-1}$, whose entries are polynomials in x with coefficients in $\mathbb{Q}[G]$; at $x = 1$ this inverts A over $\mathbb{Q}[G]$. In particular $x \mapsto T_{M(x)}$ is a polynomial, hence continuously differentiable, path of invertible operators. No identities $PU = U$, $VP = V$ or $V = U^*$ are required of the lifts. □

4 The trace calculation

Let $\mathcal{A}_n$ be the operator-norm closure of $\{T_a : a \in \mathrm{M}_n(\mathbb{C}[G])\}$, a unital C^*-algebra. Three facts about τ_n from (1.1) are used.

(T1) $|\tau_n(T)| \leq n\|T\|$ and $\tau_n(T^*) = \overline{\tau_n(T)}$, directly from $\tau_n(T) = \sum_i \langle Te_i, e_i\rangle$.

(T2) $\tau_n(ab) = \tau_n(ba)$ for $a, b \in \mathrm{M}_n(\mathbb{C}[G])$: both sides equal $\sum_{i,j}\sum_g (a_{ij})_g (b_{ji})_{g^{-1}}$.

(T3) By (T1), τ_n is norm-continuous, so (T2) extends from the dense subalgebra to all of $\mathcal{A}_n$.

Lemma 4.1. *For a continuously differentiable path $M : [0,1] \to \mathcal{A}_n$ of invertible elements,*

$$\frac{d}{dx}\tau_n\big(\log(M(x)^*M(x))\big) = 2\,\mathrm{Re}\,\tau_n\big(M'(x)M(x)^{-1}\big).$$

This is [4, Lemma 2] with the unnormalized trace; it is used only in Lemma 4.2.

Proof. Plan. Expand $\log N$, $N = M^*M$, in a norm-convergent power series, differentiate termwise, and use cyclicity to reach $\tau_n(N^{-1}N')$.

Step 1 (bounds). Inversion is continuous, so $\|M(x)\|$ and $\|M(x)^{-1}\|$ are bounded on $[0,1]$. Since $\langle N\xi,\xi\rangle = \|M\xi\|^2 \geq \|M^{-1}\|^{-2}\|\xi\|^2$, there are constants $0 < \delta < c$ with $\delta I \leq N(x) \leq cI$ for all x.

Step 2 (series). Put $\Phi = I - N/c$ (the source's W). Then $0 \leq \Phi \leq \eta I$ with $\eta = 1 - \delta/c < 1$, so $\|\Phi\| \leq \eta$, and functional calculus applied to $N = c(I - \Phi)$ gives, with norm convergence uniform in x,

$$\log N = (\log c)I - \sum_{j\geq 1} \frac{\Phi^j}{j}.$$

Step 3 (termwise differentiation). Here $N' = M'^*M + M^*M'$ and $\Phi' = -N'/c$ are continuous. The product rule gives $(\Phi^j)' = \sum_{i=0}^{j-1} \Phi^i\Phi'\Phi^{j-1-i}$, hence

$$\left\| \frac{d}{dx}\frac{\Phi^j}{j} \right\| \leq \eta^{j-1} \sup_{y\in[0,1]} \|\Phi'(y)\|.$$

The series of derivatives therefore converges uniformly, so $\log N$ is continuously differentiable with termwise derivative. By (T1), τ_n commutes with the sum and the derivative.

Step 4 (cyclicity). By (T3) each of the j summands of $(\Phi^j)'$ has trace $\tau_n(\Phi^{j-1}\Phi')$; commutation of N and N' is not assumed. Therefore

$$\frac{d}{dx}\tau_n(\log N) = -\sum_{j\geq 1} \tau_n(\Phi^{j-1}\Phi') = \frac{1}{c}\,\tau_n\Big(\sum_{j\geq 1} \Phi^{j-1}N'\Big) = \tau_n(N^{-1}N'),$$

because $\sum_{j\geq 1} \Phi^{j-1} = (I-\Phi)^{-1} = cN^{-1}$.

Step 5 (two summands). Use $N^{-1} = M^{-1}(M^*)^{-1}$. First, $\tau_n(N^{-1}M^*M') = \tau_n(M^{-1}M') = \tau_n(M'M^{-1})$. Second, by cyclicity and (T1),

$$\tau_n(N^{-1}M'^*M) = \tau_n(MN^{-1}M'^*) = \tau_n\big((M^*)^{-1}M'^*\big) = \tau_n\big((M'M^{-1})^*\big) = \overline{\tau_n(M'M^{-1})}.$$

The sum is $2\,\mathrm{Re}\,\tau_n(M'M^{-1})$. □

Lemma 4.2. *Let $r, m \geq 1$ and let $B \in \mathcal{A}_{r+m}$ be invertible. Let α, β have respective sizes $m \times r$ and $r \times m$ over the operator-norm closure of $\mathbb{C}[G]$. For $x \in [0,1]$, let L_x, R_x be lower and upper block shears with diagonal blocks I_r, I_m and off-diagonal blocks $x\alpha, x\beta$, respectively. Then $\tau_{r+m}(\log((L_xBR_x)^*(L_xBR_x)))$ is constant.*

(The source calls the blocks S, T; we rename them since S is the set of Proposition 2.2.) Used in Proposition 4.3 with $\alpha = U/q$, $\beta = V/q$.

Proof. Write $M = L_xBR_x$, a polynomial path with $M^{-1} = R_x^{-1}B^{-1}L_x^{-1}$, the shear inverses negating the off-diagonal blocks. From $M' = L_x'BR_x + L_xBR_x'$,

$$M'M^{-1} = L_x'L_x^{-1} + (L_xB)(R_x'R_x^{-1})(L_xB)^{-1}.$$

Direct multiplication gives

$$L'_x L_x^{-1} = \begin{pmatrix} 0 & 0 \\ \alpha & 0 \end{pmatrix}, \qquad R'_x R_x^{-1} = \begin{pmatrix} 0 & \beta \\ 0 & 0 \end{pmatrix}.$$

Both have zero diagonal entries, hence trace zero by (1.1). By (T3) the conjugate $(L_x B)(R'_x R_x^{-1})(L_x B)^{-1}$ also has trace zero. So $\tau_{r+m}(M'M^{-1}) = 0$, and Lemma 4.1 gives zero derivative. □

Proposition 4.3. *Let G be a discrete group, $m, r \geq 1$, and let $P = P^* = P^2 \in \mathrm{M}_m(\mathbb{Q}[G])$. Suppose an odd $D > 0$ clears its denominators and $\bar{P} = uv$ with $u \in \mathrm{M}_{m,r}(\mathbb{F}_2[G])$ and $v \in \mathrm{M}_{r,m}(\mathbb{F}_2[G])$. For every $q = 2^k > 1$ congruent to one modulo D, the matrix A of Lemma 3.2 satisfies*

$$\tau_{r+m}(\log(T_A^* T_A)) = 2(r - \tau_m(P)) \log q, \tag{4.1}$$

$$\log \det_{\mathcal{N}(G)}(T_A) = (r - \tau_m(P)) \log q. \tag{4.2}$$

It is integral and rationally invertible, and its positive-square spectral measure has support in a compact subinterval of $(0, \infty)$.

Proof. Step 1. Integrality and the inverse over $\mathbb{Q}[G]$ are Lemma 3.2. Hence T_A has the bounded inverse $T_{A^{-1}}$, and $\|T_A^{-1}\|^{-2} I \leq T_A^* T_A \leq \|T_A\|^2 I$ gives the support statement.

Step 2 (shear invariance). Apply Lemma 4.2 to T_B (invertible by Step 2 of Lemma 3.2) with $\alpha = T_{U/q}$, $\beta = T_{V/q}$. By (3.2) the path is $T_{M(x)}$, with $M(1) = A$ and $M(0) = B$. So $\tau_{r+m}(\log(T_A^* T_A)) = \tau_{r+m}(\log(T_B^* T_B))$.

Step 3 (value at $x = 0$). Since $P = P^* = P^2$, $J^* = J$ and $J^2 = q^{-2}P + (I_m - P)$, so

$$B^* B = \mathrm{diag}(q^2 I_r,\ q^{-2}P + I_m - P).$$

For a projection P and real s, $\exp(sP) = I_m + (e^s - 1)P$; with $s = -2\log q$ this is $q^{-2}P + I_m - P$. Hence

$$\log(B^* B) = 2 \log q \ \mathrm{diag}(I_r, -P),$$

with trace $2(r - \tau_m(P)) \log q$. This proves (4.1); half of it is (4.2) by (1.3).

Step 4. Dividing (4.1) by $r + m$, as in (1.3),

$$\int_{(0,\infty)} \log t \, d\mu_A(t) = \frac{2(r - \tau_m(P)) \log q}{r + m}. \tag{4.3}$$

□

5 Assembly and exact determinant

Proof of Theorem 1.1. Take K_2, H_2, a_2, b_2 from Theorem 1.2 and put $d = [K_2 : \mathbb{F}_2] \geq 1$. Lemma 2.1 gives $X, Y \in \mathrm{M}_d(\mathbb{F}_2[H_2])$ with $XY = I_d \neq YX$. Proposition 2.2 then gives $G_D = (\bigoplus_{H_2} C_3) \rtimes H_2$ and an orthogonal projection P of size $d + 1$ with

$$\tau_{d+1}(P) = d + \theta, \qquad \theta = \frac{1}{3}\left(\frac{2}{3}\right)^{|S|} > 0, \qquad D = 3^{1+|S|},$$

where the odd integer D clears the coefficients of P and $\bar{P}$ factors through d columns. Choose $k \geq 1$ with $q = 2^k \equiv 1 \pmod{D}$. Proposition 4.3 with $r = d$, $m = d + 1$ produces $A \in \mathrm{M}_{2d+1}(\mathbb{Z}[G_D])$, invertible over $\mathbb{Q}[G_D]$, with

$$\log \det_{\mathcal{N}(G_D)}(T_A) = (d - (d + \theta)) \log q = -\theta \log q.$$

Since $q > 1$ and $\theta > 0$,

$$0 < \det_{\mathcal{N}(G_D)}(T_A) = q^{-\theta} < 1. \tag{5.1}$$

With $n = 2d + 1$, Equation (4.3) reads

$$\int_{(0,\infty)} \log t \, d\mu_A(t) = -\frac{2\theta \log q}{2d+1} < 0, \tag{5.2}$$

finite by the support statement of Proposition 4.3. Finally, G_D is generated by a finite generating set of H_2 together with z_1, since $z_h = hz_1h^{-1}$. □

Remarks. The group G_D differs from the companion's H_2 and contains elements of order three. The spectrum of $T_A^*T_A$ avoids zero, so this is a failure of the lower bound, not of determinant class. The source adds an external consequence, not used above: since $T_{A^*A} = T_A^*T_A$ is positive, (1.3) gives $\log\det T_{A^*A} = \tau_n(\log(T_A^*T_A)) = -2\theta\log q < 0$, so applying the sofic determinant bound [3, Theorem 5] to the positive integral matrix A^*A shows that G_D is not sofic.

6 Verification checklist

1. **The external input.** Theorem 1.2 is the only ingredient not proved here, and everything is conditional on it (see the boxed note).
2. **The trace gap.** $psp = 0$ and $[1]p = \theta$: Proposition 2.2, Steps 3–4. Check that sps^{-1} contains p_s and that the labels $\{1\} \cup S$ are distinct.
3. **Factorization through d columns.** $\bar{p}h_0^{-1}C_{ij}\bar{p} = \bar{p}$ and $uv = \bar{P}$: Proposition 2.2, Steps 5–6, using all three identities in (2.1).
4. **Lifting.** $W^k = 0$ and $QZ = E$: Lemma 3.1, Steps 3–4.
5. **Integrality of A.** Lemma 3.2, Step 1; it needs both $q \equiv 1 \pmod{D}$ and $UV \equiv E \pmod{q}$.
6. **$A = L_1BR_1$.** The block product (3.2).
7. **Shear invariance.** Termwise differentiation and cyclicity (Lemma 4.1, Steps 3–5); vanishing diagonal of $L_x'L_x^{-1}$, $R_x'R_x^{-1}$ (Lemma 4.2).
8. **Final value and sign.** $\log(B^*B) = 2\log q \, \mathrm{diag}(I_r, -P)$ (Proposition 4.3, Step 3), then $r - \tau_m(P) = -\theta$ (Section 5).

References

[1] Pierre de la Harpe and Georges Skandalis. Déterminant associé à une trace sur une algèbre de Banach. *Annales de l'Institut Fourier*, 34(1):241–260, 1984.

[2] Warren Dicks and Thomas Schick. The spectral measure of certain elements of the complex group ring of a wreath product. *Geometriae Dedicata*, 93:121–137, 2002.

[3] Gábor Elek and Endre Szabó. Hyperlinearity, essentially free actions and L^2-invariants. The sofic property. *Mathematische Annalen*, 332:421–441, 2005.

[4] Bent Fuglede and Richard V. Kadison. Determinant theory in finite factors. *Annals of Mathematics*, 55(3):520–530, 1952.

[5] Holger Kammeyer. A counterexample to the determinant approximation conjecture. arXiv:2609.15567v1, 2026. Version 1, September 14, 2026.

[6] Franz Lehner, Markus Neuhauser, and Wolfgang Woess. On the spectrum of lamplighter groups and percolation clusters. *Mathematische Annalen*, 342(1):69–89, 2008.

[7] Wolfgang Lück. *L^2-Invariants: Theory and Applications to Geometry and K-Theory*, volume 44 of *Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge*. Springer, 2002.

[8] OpenAI. A Counterexample to Kaplansky's Direct-Finiteness Conjecture in Characteristic Two. OpenAI Math Release preprint OAI:A-Counterexample-to-Kaplanskys-Direct-Finiteness-Conjecture-in-Characteristic-Two-September-23-2026, 2026. Theorem 1.1 and Section 6.

[9] Thomas Schick. L^2-determinant class and approximation of L^2-Betti numbers. *Transactions of the American Mathematical Society*, 353(8):3247–3265, 2001. With the appended erratum in arXiv:math/9807032v2.